

Machine learning assisted prediction and optimization of mechanical properties for laser powder bed fusion of Ti6Al4V alloy

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ABSTRACT

Due to the complex physical metallurgy phenomena and enormous parameter combination, the traditional trial-and-error method makes the microstructure tailoring of laser additive manufactured (LAM) for exceptional performance still a major challenge. Here, we presented a machine learning-based model to facilitate the parameter optimization and microstructure tailoring of laser powder bed fused (L-PBF) Ti6Al4V alloy with enhanced strength-ductility synergy. Initially, a database was constructed based on the 173 data sets from 31 related literature, with an in-depth analysis of key parameters such as laser power, laser speed, and powder size using the Pearson correlation coefficient (PCC). K-Means clustering was integrated into the Clustering Integrated Regression Model (CIRM), enhancing the cohesion of similar data groups based on process parameters. This strategic clustering significantly increases the precision of tailored predictive models for each group, markedly improving overall prediction accuracy. Additionally, combined with non-dominated sorting Genetic Algorithm II (NSGA-II), the CIRM model ensures rapid optimization and achieves the balance between strength and ductility during multi-objective optimization. L-PBF experiments, based on optimized parameters provided by the NSGA-II model, demonstrated an excellent combination of strength and ductility, compared to existing literature. Moreover, the Shapley additive explanation (SHAP) was introduced to interpret the prediction model, which indicates that adjusting the grain size distribution of martensite through laser-related parameters is critical for simultaneously enhancing strength and ductility. Essentially, our work provides a robust approach for the accurate prediction and multi-objective optimization of mechanical properties in LAM metallic materials.

1. Introduction

As an emerging manufacturing technology, laser additive manufacturing (LAM) is capable of directly fabricating near-net-shaped components in a layer-by-layer fashion from CAD data files. This capability has attracted significant attention for LAM methods such as laser powder bed fusion (L-PBF) [1,2] and laser-directed energy deposition (L-DED) [3], which provide favorable opportunities for manufacturing geometrically complex and high-performance alloys [4,5]. Due to its high degree of design freedom [4], LAM has become a part of product development in fields such as aerospace and medical industries. One notable example is Ti6Al4V in LAM, which is suitable for jet engines, aerospace components, and medical equipment [6–9]. However, the application of LAM currently faces bottlenecks and challenges, primarily

due to a series of problems including intrinsic metallurgical defects, microstructure modulation, and insufficient mechanical properties [10–12]. These problems stem not only from the extensive and complex process parameters involved in laser additive manufacturing but also from the complexity of its forming process. This process encompasses numerous intricate physical metallurgy phenomena during the layer-by-layer rapid melting and solidification, which directly impact the performance of the resulting alloy material [13,14].

The focus in LAM for high-performance alloys resides in microstructure control and mechanical properties improvement. The trial-and-error method is commonly employed to find the optimal parameter combination within a narrow range [15]. Researchers have employed the trial-and-error method to explore the influence of four process parameters related to volumetric energy density. Nonetheless,

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the results show significant variations in alloy microstructure and properties even when the same volumetric energy density is applied [1]. Post-heat treatment [16,17] and external magnetic fields [18] can also be used to manipulate the structure and properties of LAM-fabricated alloys. However, these approaches have limitations as they involve additional techniques that complicate the overall process, requiring substantial time and financial investment. Therefore, our objective remains to optimize the mechanical properties through the manipulation of process parameters.

In the last decade, machine learning (ML) has been employed as a promising approach in materials manufacturing to facilitate the screening of high-performance alloys because of its ability to explore relationships behind existing experimental data for prediction and optimization [19–22]. There has been increasing attention paid to optimizing alloy composition, process parameters, metallurgical defects, properties, and process monitoring using machine learning methods. Gao et al. [23] established a quantitative relation among “composition-processes-properties” in L-PBF Al-Si-(Mg) alloys to design alloys for a maximum comprehensive mechanical property. Tapia et al. [24] utilized Gaussian process regression (GPR) to predict the depth of single-pass melt pool and part density, thus addressing process parameter optimization problems using a small range of input data. Chen et al. [25] proposed probabilistic physics-guided neural networks for analyzing the probabilistic fatigue properties of L-PBF Ti6Al4V by considering the effect of process parameters. Liu et al. [26] developed a machine learning method based on GPR to determine the optimized processing window and found a new and much larger optimized L-PBF processing window for fabricating fully dense AlSi10Mg samples. Akbari et al. [27] introduced a comprehensive framework for benchmarking ML of melt pool characterization. They introduced physics-aware LAM featurization, versatile ML models, and evaluation metrics to create a comprehensive learning framework for melt pool defect and geometry prediction.

Mechanical properties are important prerequisites for Ti6Al4V alloy in engineering applications. However, Ti6Al4V alloy fabricated by L-PBF tends to exhibit low elongation to fracture and high strength, due to the nearly full martensite microstructure under an extremely high cooling rate. Therefore, it has become a challenge to balance elongation to fracture and strength. However, comprehensive studies on applying machine learning in LAM from prediction of mechanical properties to optimization of process parameters are still scarce. To achieve this, it is required to establish an accurate and fast prediction for the complicated LAM process by leveraging a limited amount of data. In addition, it further requires an inverse parameter search of optimal solutions in the process parameter space for the objectives of mechanical properties based on forward prediction results. These two main parts make the optimization of LAM process parameters challenging, which motivates this work to explore and even achieve this goal. LAM processes involve complex non-equilibrium and multi-physics phenomena, which are influenced by a range of laser-related and powder-related parameters. In addition to laser power and scanning speed, factors such as laser spot size and powder particle size can significantly impact the LAM process, leading to changes in laser energy input density and laser beam reflection. These factors play a crucial role in determining rapid melting and solidification parameters, including high temperatures and cooling rates in the melt pool, thereby affecting microstructure and mechanical properties [28]. The gauge length is used to calculate the strain that occurs in the sample during the tensile test, expressed as $(\Delta L/L_0)$, where ΔL is the change in gauge length and L_0 is the original gauge length. Accurate strain calculation is crucial for evaluating the material's mechanical properties. The formula for stress (σ) is given by $(\sigma = F/A)$, where F is the applied force and A is the gauge area. The gauge area directly affects the stress calculation, which in turn influences the assessment of material strength. This research specifically focuses on the L-PBF fabrication process of the Ti6Al4V alloy.

In the present work, aiming at the L-PBF of Ti6Al4V alloy, an

integrated multi-objective optimization framework incorporating a prediction surrogate model of mechanical properties based on the machine learning concept [29] was developed. In the training and prediction step, the dataset is first divided into clusters using a clustering algorithm [30]. The optimal ML model for each cluster is then selected based on the prediction performance of varied ML models to achieve the highest local accuracy, resulting in the best global prediction result. Furthermore, the optimal set of process parameters is obtained using a non-dominated Pareto-based multi-objective optimization method. This framework enables high-efficiency and high-accuracy optimization of process parameters in LAM, and it is capable of being deployed in a large range of potential intervals in additive manufacturing.

2. Methods

The design methodology of process parameters used follows the “goal-means” material design philosophy [31], which starts by defining the target properties required by the application, and then the target process parameters are subsequently designed to obtain the desired properties, according to the algorithm, metallurgical, and mechanical principles. The objective of this work is to leverage existing data for predictive properties and process optimization. The usable data was collected from an existing literature database. A framework containing two main parts of prediction and optimization is established. The prediction module is the clustering integrated regression model (CIRM) based on the K-Means algorithm and machine learning regression models. The optimization module is Non-Dominated Sorting Genetic Algorithm II (NSGA2) using CIRM as the surrogate model. The optimal process parameter combinations for the alloys with the best mechanical properties optimized by the framework are verified through experiments.

2.1. Dataset preparation

The process parameters considered in this study are incorporated as inputs into the machine learning model. These parameters include laser power, scanning speed, hatch distance, layer thickness of the powder bed, powder size, laser spot size of the equipment, as well as the gauge length and gauge area during tensile tests. The eight inputs are utilized to generate three outputs: ultimate tensile strength (UTS), yield strength (YS), and elongation to fracture (EF). Published articles on L-PBF of Ti6Al4V were gathered to construct the database for the predictive machine learning model. The data collection focuses on the parametric study of mechanical properties, emphasizing the establishment of the correlation between laser and powder-related parameters and tensile test properties. The database includes a total of 173 high-quality parameter combinations from 31 papers (see the supplementary file). The statistics with the parameter range are shown in Table 1.

As shown in Table 1, there are dimensional differences among the ranges of process parameters. In order to mitigate the impact of these differences on the model, data pre-processing is necessary before using it for training the model [32]. The z-score normalization method [33] is adopted for preprocessing the process parameters, which is shown in Eq. 1. Z-score normalization retains the original units of the data, which is crucial for eight process parameters with varying units and numerical ranges. Moreover, Certain algorithms employed in the text, such as support vector regression and random forests, are sensitive to the scale of input data. Z-score normalization can reduce the variance in these algorithms during training, potentially enhancing the model's stability and accuracy.

$$Z = \frac{x - \mu}{\sigma} \quad (1)$$

where μ and σ are the mean and standard deviation of the process parameter x . The same approach was employed for the data pre-processing of mechanical properties.

Table 1

Input and output ranges of the various parameters in the total dataset.

	Inputs and outputs	Minimum	Maximum	Mean	Standard deviation
Inputs	Powder size (μm)	27.0	70.0	35.79	7.50
	Laser spot (μm)	40.0	120.0	80.24	14.19
	Laser power (W)	50.0	500.0	224.47	110.60
	Scanning speed (mm/s)	220.0	2000.0	926.40	362.19
	Hatch distance (μm)	50.0	180.0	90.63	22.69
	Layer thickness (μm)	20.0	80.0	41.85	16.01
	Gauge length (mm)	3.0	81.0	23.86	9.97
	Gauge area (mm^2)	2.0	78.5	12.45	13.95
	Ultimate tensile strength (MPa)	794.0	1484.9	1122.73	130.73
Outputs	Yield strength (MPa)	643.0	1366.3	1005.20	127.41
	Elongation to fracture (%)	1.3	18.5	8.08	3.51

2.2. Analysis of process parameters

After data collection and pre-processing, the analysis of Pearson correlation coefficient (PCC) [34] is conducted to understand the correlation among these process parameters, which is calculated according to the Eq. 2.

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{E(X - \mu_X)(Y - \mu_Y)}{\sigma_X \sigma_Y} \quad (2)$$

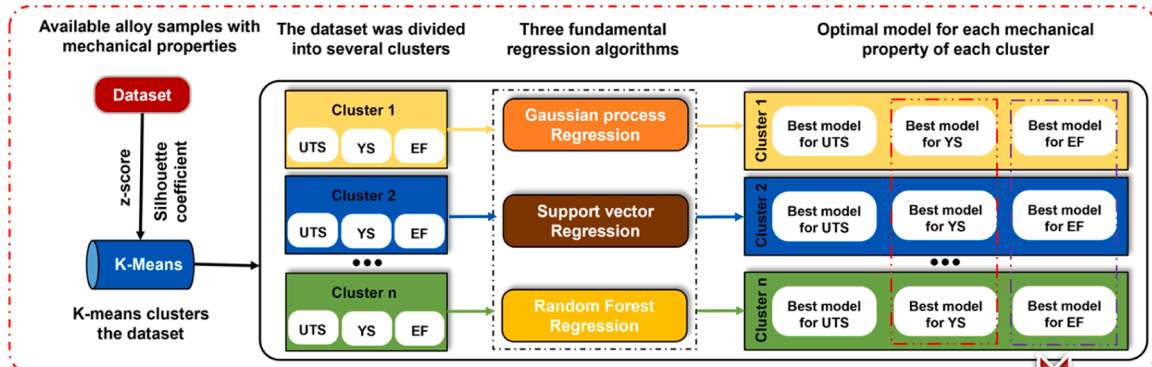
where ρ_{XY} denotes the correlation coefficient, X and Y are any two different characteristics, cov is the covariance, μ and σ are the mean and standard deviation.

To further elucidate the relationship between process parameters and mechanical properties, the SHAP method [35] is introduced to explain the mechanical property prediction model. It can help us understand how each process parameter affects the output of the model and enables us to evaluate the importance of features (i.e., process parameters) in complex machine learning models based on SHAP values.

2.3. Construction of prediction model for mechanical properties

In the prediction model, the mechanical properties (UTS, YS, and EF) were set as the targets of the prediction which was achieved using the CIRM method in this study. The implementation process is shown in Fig. 1. The K-Means algorithm clusters the normalized dataset, and the number of clusters is determined by evaluating the silhouette coefficient. Each cluster is individually trained using fundamental algorithms (GPR, SVR, and RF), and the optimal model is selected for each

Clustering Integrated Regression Model (CIRM)



Non-dominated Sorting Genetic Algorithm - II (NSGA2)

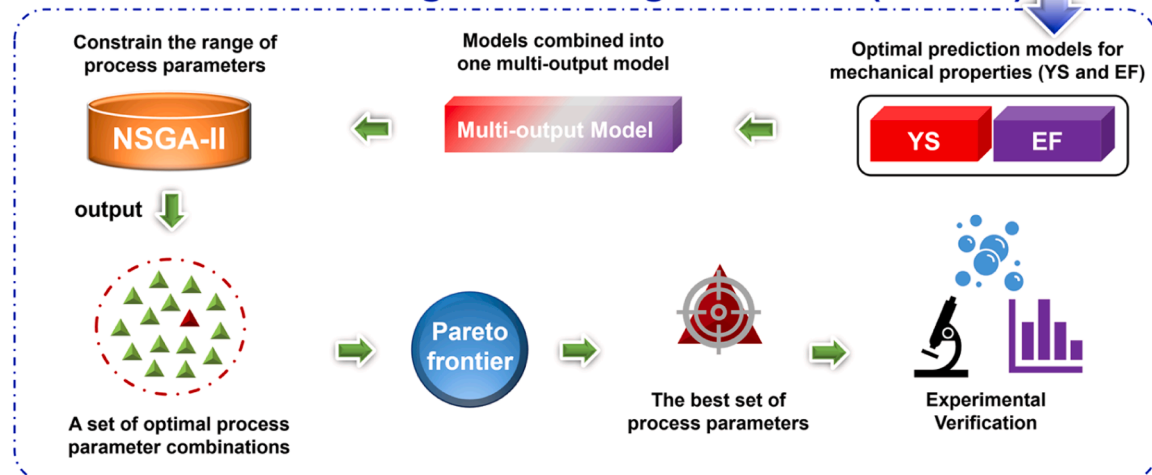


Fig. 1. Schematic diagram of the design method and model validation.

mechanical property of each cluster. These optimal models from each cluster are then integrated to form the overall CIRM model.

K-Means is an unsupervised clustering algorithm [36]. In this study, K-Means partitioned data into different clusters based on the similarity of their mean and standard deviation of process parameters. Generally, different clusters have significantly different mean and standard deviation values for process parameters. Thus, it can identify alloys with similar combinations of process parameters over an appropriate spatial range. The K-Means algorithm is a distance-based clustering algorithm. Given a sample dataset $Q = \{x_1, x_2, \dots, x_n\}$, where x_i denote the i^{th} sample, K-Means algorithm finds the k cluster centers $\{o_1, o_2, \dots, o_k\}$ that minimizes the sum of the Cosine distance between each sample x_i and its nearest cluster center o_i . The Cosine distance between two vectors is presented in Eq. 3.

$$Dis(x_i, o_k) = \frac{\sum_{j=1}^n x_{ij} \bullet o_{jk}}{\sqrt{\sum_{j=1}^n x_{ij}^2} \bullet \sqrt{\sum_{j=1}^n o_{jk}^2}} \quad (3)$$

Where $Dis(x_i, o_k)$ denotes the Cosine distance between x_i and o_k , x_{ij} is the j^{th} feature value of the samples i , and o_{jk} is the j^{th} dimension value of o_k . The algorithm separates all samples into k compact and independent clusters based on the principle that the higher the similarity between two samples, the closer their distance. In addition, The t-Distributed Stochastic Neighbor Embedding (t-SNE) method [37] is employed to visualize the clustering results in a two-dimensional space. t-SNE is a non-linear dimensionality reduction algorithm that maps high-dimensional data onto a two-dimensional plane while preserving local structure, enabling better visualization, and understanding of data.

In order to set the rational value of k , according to the feature dimension, the range of k is evaluated from 2 to 8 one by one using the silhouette coefficient [38] method, which determines the most reasonable number of clusters. The silhouette coefficient formula is shown in Eq. 4. Where $a(i)$ represents the mean of the distance between sample i and each sample in the cluster, $b(i)$ represents the mean of the distance from sample i to every other sample within the nearest cluster.

$$S(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \quad (4)$$

After dividing the dataset into the most appropriate number of clusters, three fundamental ML algorithms (GPR [39], SVR [40], and RF [41]) are employed in each cluster for the construction of CIRM. The optimal model is determined for each cluster by the Squared Correlation Coefficient (R^2), which describes the fitness between the stretching mechanism and the candidate model, as shown in Eq. 5.

$$R^2 = 1 - \frac{\sum_{i=1}^n (Y_i - \bar{Y}_i)^2}{\sum_{i=1}^n (Y_i - \bar{Y}_i)^2} \quad (5)$$

where Y_i is the sequence of true values, y_i is the sequence of predicted values, and $\bar{Y}_i$ denotes the average true value.

The performances of the models are further evaluated using Mean Absolute Error (MAE) to verify the accuracy of the model (shown in Eq. 6).

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - Y_i| \quad (6)$$

where n is the number of samples, y_i and Y_i represent the predicted value and true value of the i^{th} sample, respectively.

To ensure no overfitting, 5-fold cross-validation is conducted on a selected model for each cluster, and the corresponding fitness value [30] is evaluated as shown in Eq. 7, where Y_i is the sequence of true values, y_i is the sequence of predicted values.

$$fitness = 1 - \frac{1}{n} \sum_{i=1}^n \frac{|y_i - Y_i|}{Y_i} \quad (7)$$

2.4. Parameters selection based on multi-objective optimization

To identify the optimal process parameters based on the desired mechanical properties, the Genetic Algorithm can be utilized for optimization in this project. The optimization objective involves both strength and elongation to fracture, which may not be attainable simultaneously. In other words, enhancing one may adversely affect the other. Thus, the NSGA2 algorithm, as a multi-objective optimization technique shown in Fig. 1, is employed to obtain a Pareto front by using the developed CIRM algorithm as a surrogate model for fast prediction. This approach effectively resolves the optimization problem associated with the conflict between strength and elongation to fracture. The Pareto front [42] represents a collection of solutions in multi-objective optimization where further improvement in one objective value cannot simultaneously improve any other objectives. Consequently, it is necessary to evaluate and identify the most optimal solution on the Pareto front by considering domain knowledge and actual conditions. In addition, among the eight different process parameters under consideration, four adjustable ones including the laser power, scanning speed, hatch distance, and layer thickness can be changed during the manufacturing process. The other three parameters including the particle size of the powder, laser spot size, and gauge of the tensile specimen, are related to the equipment and materials, which are not adjustable in the experiment. The constraints imposed by the actual laboratory equipment are set in the NSGA2 algorithm and the ranges of process parameters are shown in Table 2.

3. Results

3.1. Correlation distribution between the process parameters

The correlations between the process parameters and the mechanical properties are shown in Fig. 2. The result shows that the correlation coefficient between any two process parameters is lower than 0.8, indicating no strong linear relationship among the process parameters. On the other hand, from the experiment perspective, these parameters are also essential in determining the successful formation of alloys within the L-PBF process. Moreover, the correlation coefficient between UTS and YS is remarkably high at 0.89, indicating a strong linear relationship. UTS represents the maximum stress that a material can withstand under tensile forces, while YS is the stress at which the material begins to deform plastically without any increase in strain. As the stress level increases, the material initially attains YS, indicating the onset of plastic deformation. Upon further stress escalation, the material ultimately fails, with the corresponding strength value being UTS. Given that both the plastic deformation and failure modes are closely associated with the strength characteristics of the material, a positive correlation is typically observed between UTS and YS. As a result, in the optimization stage, selecting one of them can decrease the number of objectives and simplify the optimization process.

Table 2

The constraint range of process parameters in the NSGA2 algorithm.

Process parameters	constraint range
Powder size (μm)	37.7
Laser spot (μm)	75
Laser power (W)	100–300
Scanning speed (mm/s)	800–2000
Hatch distance (μm)	30–150
Layer thickness (μm)	30
Gauge length (mm)	8
Gauge area (mm^2)	2

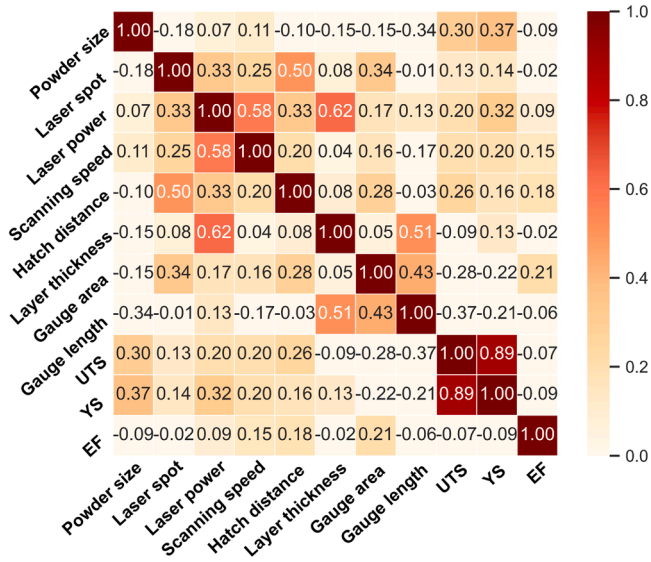


Fig. 2. Pearson correlation coefficients of process parameters.

3.2. K-Means algorithm clustering results

The value of k is determined by the silhouette coefficient. Through the computation results, it can be concluded that the highest value of the silhouette coefficient is obtained when k is 3 (in Fig. 3(a)). As a result, three distinct clusters are identified, with the corresponding number of samples in each cluster displayed in Fig. 3(b). To facilitate visual inspection of the distribution patterns, the t-SNE is further applied to reduce the high-dimensional clustering results to a two-dimensional image. The number of samples for clusters 1, 2, and 3 are 23, 72, and 78, respectively. The mean and standard deviation of each cluster are shown in Table 3. The clustering results of the data are mainly based on the similarity of the numerical value of process parameters by checking the mean and standard deviation of each cluster statistically. To obtain the optimal models for each cluster, each cluster is divided into training and test sets with a ratio of 4:1.

3.3. Optimal model selected for the clustering algorithm

The optimal model for each mechanical property of each cluster is selected by comparing R^2 of the test set, and the results are shown in Table 4 below.

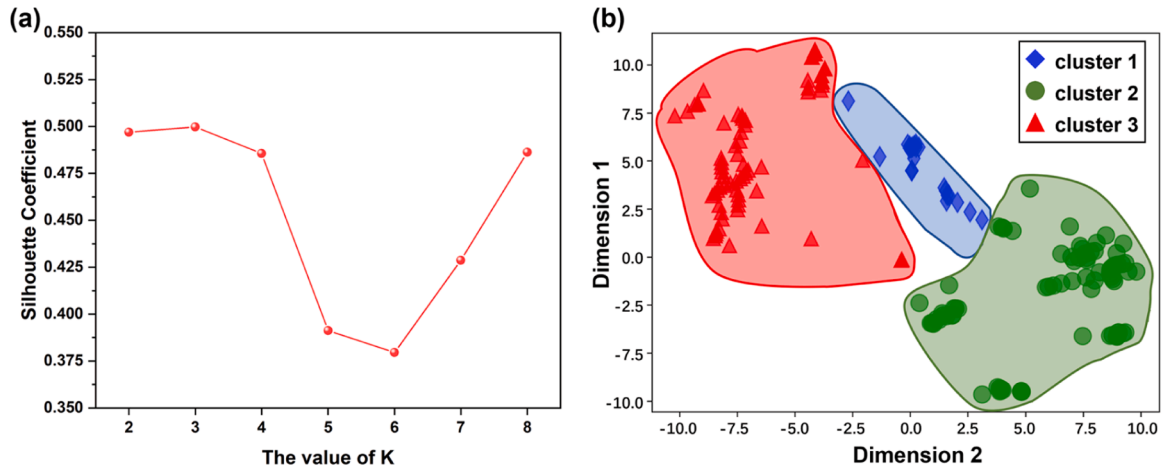


Fig. 3. Determination of the value of k in the K-Means algorithm and the results of K-Means clustering: (a) value of the silhouette coefficient when k is taken from 2 to 8; (b) two-dimensional visualization of the clustering result.

Models with the maximum R^2 values for each mechanical property are selected in each cluster (as shown in Fig. 4), and the results showed that the chosen models are precisely the same for all three mechanical properties in each cluster. Therefore, only one model is needed in each cluster. Models for different clusters are then integrated and combined with the K-means algorithm to form the cluster-integrated regression model (CIRM) for the prediction of each objective. To test the performance of the CIRM for each mechanical property, 5-fold cross-validation and average fitness values are carried out, and they strongly demonstrated the effectiveness of the model (shown in Fig. 5). The error bars shown in Fig. 5 represent the maximum and minimum fitness values for each objective.

3.4. Effect of K-Means on the prediction model

To demonstrate the benefits of such clustering for improving the accuracy of the final prediction results, three separate regression algorithms are directly utilized to predict targets without using clustering, namely GPR, SVR, and RF, and they are compared with the CIRM algorithm. As shown in Fig. 6, the mechanical properties are measured on the horizontal axis (true values) and predicted on the vertical axis (predicted values). A closer diagonal alignment along the 45-degree line indicates better prediction performance.

By examining the figure (shown in Fig. 6), it becomes evident that the CIRM consistently outperforms the other three algorithms in terms of accuracy and degree of variation for all three mechanical properties. The CIRM not only provides more accurate predictions but also exhibits a lower degree of deviation in prediction, indicating a lower overfitting. These findings highlight the advantages of utilizing the clustering algorithm in improving the precision of the final prediction results.

3.5. Multi-objective optimization

The curve formed by optimal nondominated data points is the Pareto front that provides a candidate of the most optimal process parameters, as shown in Fig. 7(a). Since a high correlation coefficient of 0.89 is found between UTS and YS in the previous correlation analysis, YS and EF are set as mechanical property parameters for multi-objective optimization for simplification. After calculating the Cosine distance between each candidate from the Pareto front and the three cluster centers, it is found that they are closest to cluster 3. Therefore, it can be concluded that the final optimization result comes from cluster 3. Fig. 7(b-d) displays the distribution of process parameters for data from the Pareto front. It should be noted that only three process parameters were incorporated instead of the original eight considering the adjustable process

Table 3

Distribution of the values of each cluster.

Parameters	Dataset		Cluster 1		Cluster 2		Cluster 3	
	Mean	Standard deviation	Mean	Standard deviation	Mean	Standard deviation	Mean	Standard deviation
Powder size (μm)	35.9	7.5	35.3	3.8	34.5	4.2	37.5	10.1
Laser spot (μm)	81.8	12.2	101.2	8.6	82.1	6.0	75.7	11.3
Laser power (W)	227.4	111.5	267.8	35.4	309.1	93.8	140.1	69.2
Scanning speed (mm/s)	916.3	349.2	1167.4	97.4	965.3	390.7	797	302.9
Hatch distance (μm)	90.1	22.9	133.0	19.2	90.1	17.0	77.4	9.1
Layer thickness (μm)	42.3	16.2	30.4	2.0	60.3	7.3	29.2	4.3
Gauge length (mm)	9.9	6.5	17.3	10.3	11.2	5.3	6.6	3.1
Gauge area (mm^2)	22.9	8.5	18.5	6.1	29.0	5.6	18.8	7.9

Table 4

Prediction model for each cluster of each mechanical property.

Algorithm		Cluster 1		Cluster 2		Cluster 3	
R ²		Train	Test	Train	Test	Train	Test
UTS	GPR	99.5 %	81.7 %	99.5 %	81.2 %	99.9 %	97.0 %
	RF	86.8 %	80.2 %	97.0 %	92.4 %	92.7 %	82.0 %
	SVR	95.3 %	91.5 %	99.1 %	81.3 %	94.7 %	80.4 %
YS	GPR	99.5 %	73.5 %	99.9 %	83.3 %	95.9 %	90.4 %
	RF	71.3 %	56.3 %	95.6 %	93.2 %	92.4 %	83.5 %
	SVR	93.6 %	86.2 %	98.6 %	83.4 %	94.7 %	82.8 %
EF	GPR	89.9 %	74.8 %	97.3 %	81.5 %	98.8 %	89.2 %
	RF	83.0 %	70.9 %	95.3 %	92.6 %	88.2 %	82.4 %
	SVR	93.0 %	87.1 %	96.4 %	81.9 %	86.5 %	82.3 %

*The values of the best sub-models in each cluster are highlighted in bold.

parameters available on the actual L-PBF equipment, as previously mentioned in Table 2. In general, a range of process parameters that often appear in the candidates is more likely to be the range of the optimal solution, and therefore we tend to choose data points in that range. It can be seen from Fig. 7 that all three process parameters of S1 are distributed in the interval with the maximum number of data points from the Pareto front, and as a result, S1 is selected as the most optimal solution. In addition, two other data points are selected to compare to S1. One sample point (S2) is selected as the process parameter distributions are comparable with S1 only except in scanning speed. We can validate the strategy of using process parameter distributions to select the most optimal solution from the Pareto front by choosing S2 as a comparison. The other sample point (S3) is selected as it has the best-predicted elongation to fracture which is often expected to improve in additive manufacturing of Ti alloy without sacrificing the strength too much. The process parameters for these three candidates, along with their predicted mechanical properties, are shown in Table 5. These candidates are used for performing experiments and validating the optimal solution. The results will be discussed in Section 4.

3.6. Effect of each process parameter on mechanical properties

To further elucidate the relationship between process parameters (features) and mechanical properties, this study calculates feature importance and evaluates their effects in predictions in the SHAP method. As shown in Fig. 8(a-c), the horizontal axis represents the SHAP value, and the vertical axis denotes different features, with each dot representing a sample. A positive/negative SHAP value for a feature indicates that the feature enhances/weakenes the mechanical property. The color of the point represents the size of the feature value. For a single feature, the wider the horizontal coverage, the greater the impact on the prediction result, indicating that the feature is more important. In other words, the mean absolute SHAP value of a feature can be regarded as its importance, as shown in Fig. 8(d-f).

When optimizing the mechanical properties of alloy manufacturing

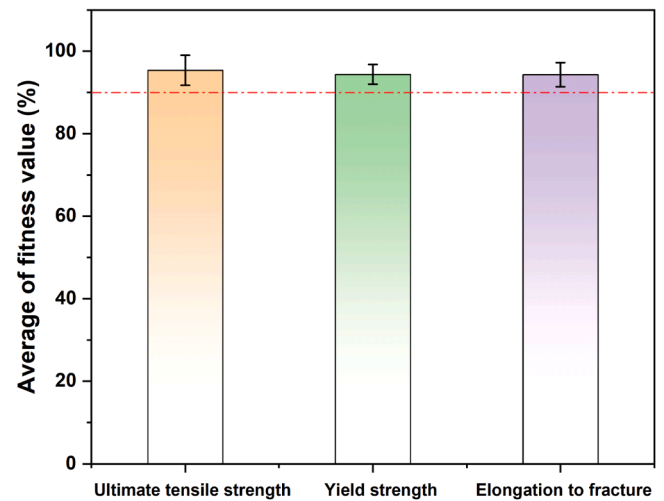


Fig. 5. Average fitness values (5-fold cross-validation) of CIRM for mechanical properties.

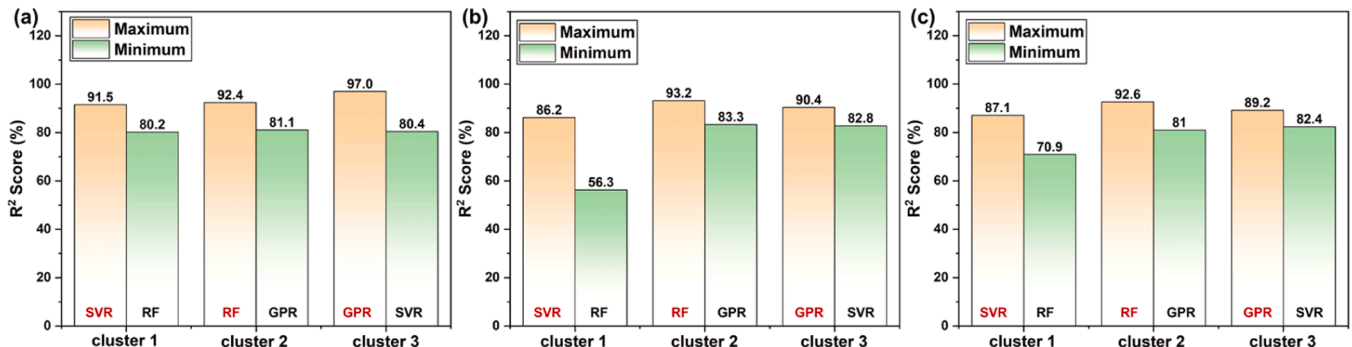


Fig. 4. Maximum and minimum values of R^2 for each cluster model of mechanical properties: (a) UTS; (b) YS; and (c) EF.

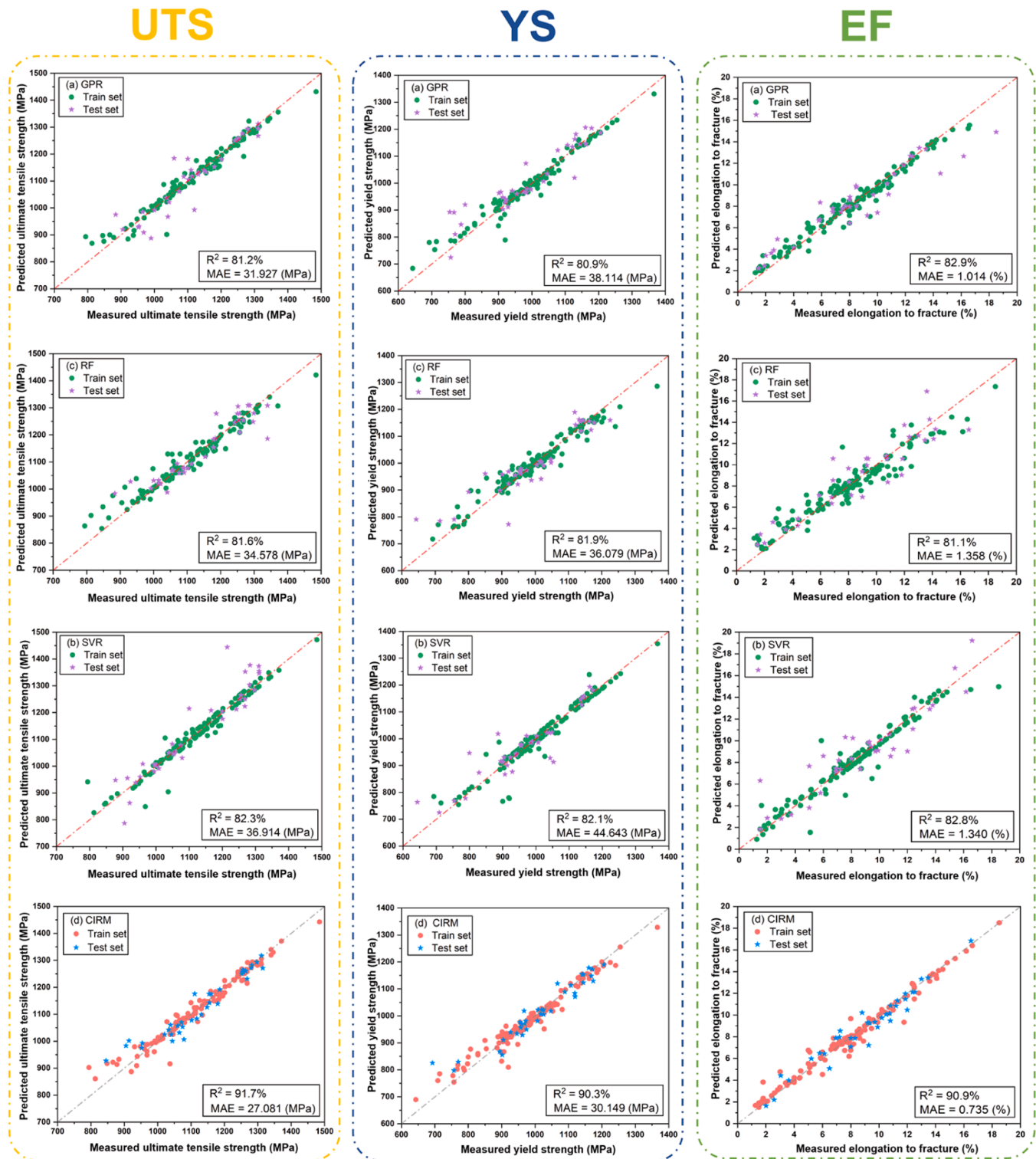


Fig. 6. Predictive performance of four machine learning models on three mechanical properties (UTS, YS, and EF) datasets: (a) GPR; (b) SVR; (c) RF; (d) CIRM.

in L-PBF, the most common adjustment is the laser power and scanning speed. Through the observation of Fig. 8(a-c) in conjunction with Fig. 9 (a-c), it is evident that in the range of 100 – 300 W, an increase in laser power leads to an enhancement in strength, while the elongation to fracture exhibits a decreasing trend. The reasons behind these results can be elucidated by referring to previous studies that the grain size of beta columnar crystal will increase with the increase of laser power, resulting in the tensile strength and yield strength of Ti6Al4V titanium

alloys tending to increase [43,44]. For the scanning speed, Fig. 9(d-f) shows intermediate range of scanning speed (1000 – 1700 mm/s) can obtain better strength and elongation to fracture, and the mechanical properties deteriorate at both low and high scanning speeds. This is because low scanning speed induces high thermal energy, leading to the explosion of the melt pool and causing high material spattering, resulting in defects that harm the mechanical properties. On the other hand, when the scanning speed is high, it causes a high rate of powder

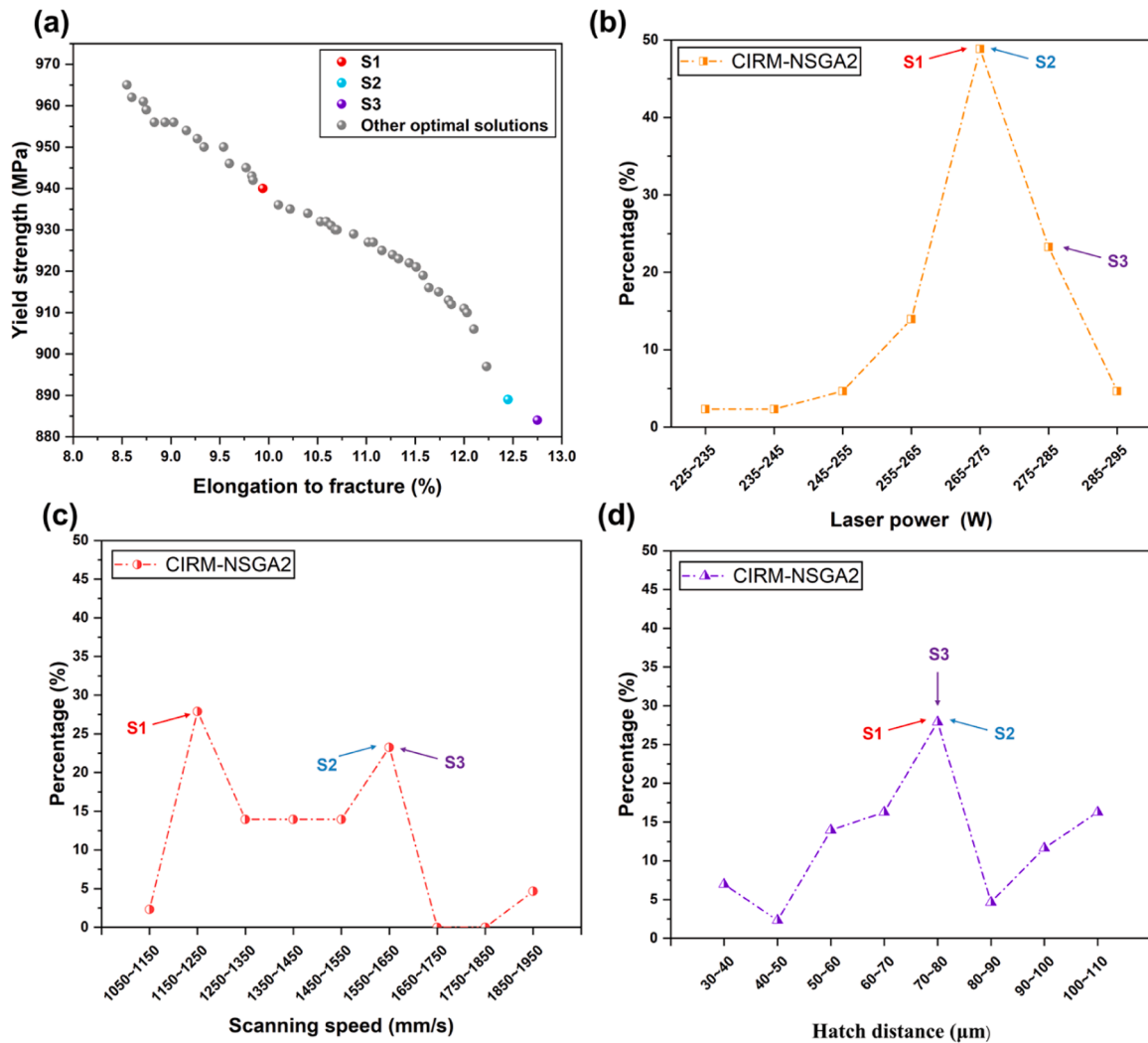


Fig. 7. (a) Evaluate the set of optimal process parameters on the combination curve of the Pareto front; Distribution of the new alloys on the Pareto front with three process parameters: (b) laser power; (c) scanning speed; (d) hatch distance.

Table 5

Process parameters predicted properties of two potential alloys evaluated by CIRM-NSGA2 model.

Sample	S1	S2	S3
Laser power (W)	266	273	276
Scanning speed (mm/s)	1218	1587	1635
Hatch distance (μm)	76	76	76
Predicted UTS (MPa)	1103	1045	1040
Predicted YS (MPa)	940	889	884
Predicted EF (%)	9.94	12.45	12.76

particle spattering and leads to insufficient melting, resulting in pores which also harm the mechanical properties [45].

The effect of hatch distance on strength and elongation to fracture is significant. In Fig. 8(a-c), it can be observed that both the strength and elongation to fracture decrease continuously as the hatch distance increases. According to related studies [46], a larger hatch distance leads to a reduction in the connection area between adjacent scan lines and a decrease in energy input, thereby increasing the potential for defect formation and resulting in poor mechanical properties. The results of the SHAP analysis in Fig. 8(a-c) show that the mechanical properties do not change significantly with respect to layer thickness, powder size, and the size of the laser spot. As shown in Fig. 8, the UTS and YS gradually decrease with the increases in the length and area of the gauge region of

the tensile specimens, which is consistent with previous research, but the EF decreases with the reduction in the gauge area.

4. Experimental validation

4.1. L-PBF experimental procedures

To validate the results of multi-objective optimization, the L-PBF experiments were conducted based on the laser processing parameters provided by the CIRM-NSGA2 model. In the validation experiments, the gas-atomized Ti6Al4V powder (AVIMETAL AM, Beijing, China) was used as the raw material for L-PBF. The powder particle possessed a spherical morphology with a size distribution by a D50 value of 37.7 μm. Cubic Ti6Al4V samples for microstructure characterization and cuboid samples for mechanical properties tests were manufactured by a commercial L-PBF machine (DMP-ProX 200, 3D Systems, USA) equipped with a 300 W fiber laser (wavelength of 1070 nm). The laser spot size is 75 μm. The chamber was prefilled with argon atmosphere to minimize oxygen contamination (< 500 ppm). Consistent with the optimization result in CIRM-NSGA2 model, the processing parameters are set as follows: S1 sample (laser power of 266 W, scanning speed of 1218 mm/s, hatch distance of 76 μm), S2 sample (laser power of 273 W, scanning speed of 1587 mm/s, hatch distance of 76 μm) and S3 sample (laser power of 276 W, scanning speed of 1635 mm/s, hatch distance of 76

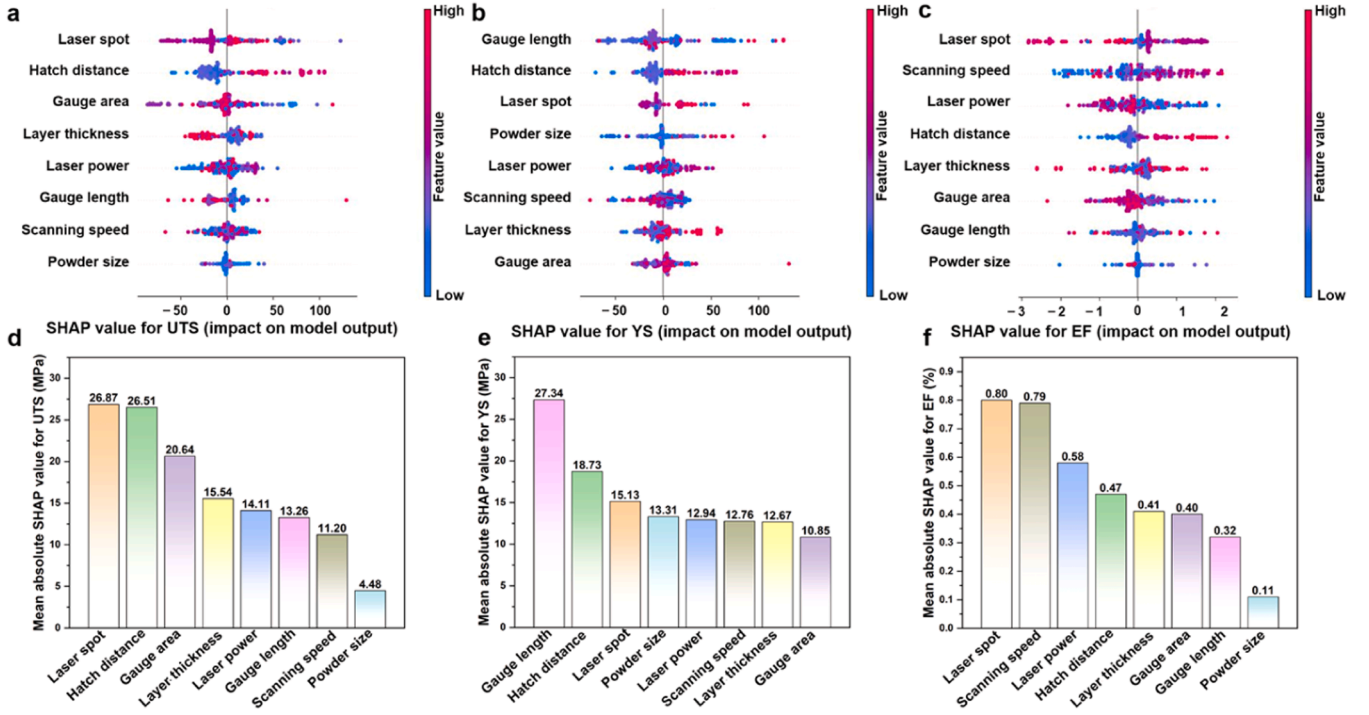


Fig. 8. Distribution of SHAP values for different samples of mechanical properties: (a) UTS, (b) YS, (c) EF; Feature importance ranked by SHAP of mechanical properties: (d) UTS, (e) YS, (f) EF.

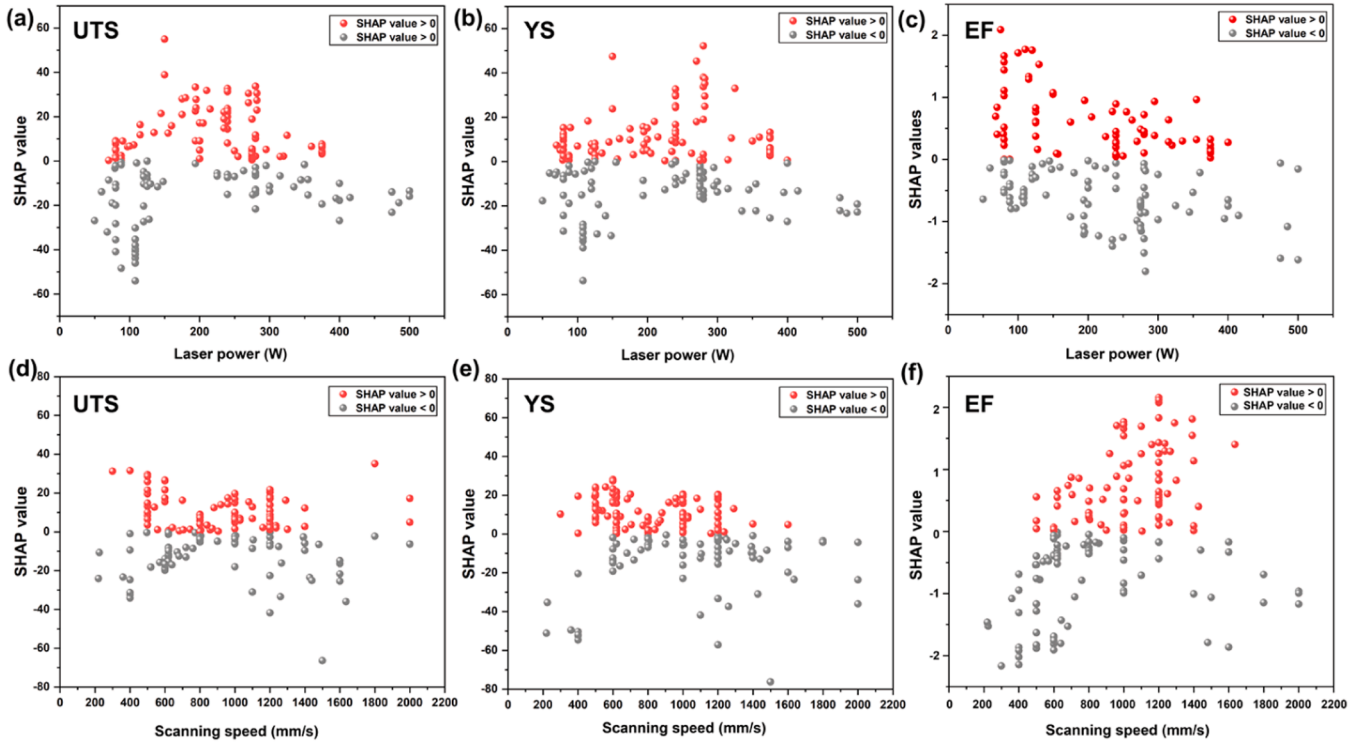


Fig. 9. Distribution of SHAP values of laser power for mechanical properties: (a) UTS, (b) YS, (c) EF; Distribution of SHAP values of scanning speed of mechanical properties: (a) UTS, (b) YS, (c) EF. The red/grey dot represents samples that have a positive/negative impact on mechanical properties.

μm). The powder layer thickness is 30 μm. Fig. 10

For the tensile test, the L-PBF Ti6Al4V samples sectioned on $x-z$ plane were conducted using Instron 5565 universal testing machine (INSTRON, USA). To ensure data accuracy, three tensile samples were repeated in the test for each processing condition. The loading direction

was aligned paralleled the building direction of the L-PBF sample at room temperature with a gauge size of $8 \times 2 \times 1 \text{ mm}^3$, and the tensile strain rate of $5 \times 10^{-4} \text{ s}^{-1}$ was utilized (in Fig. 11). Before the start of testing, our specimens were polished with 600-grit sandpaper to remove the marks left by wire cutting. The microstructure was investigated by

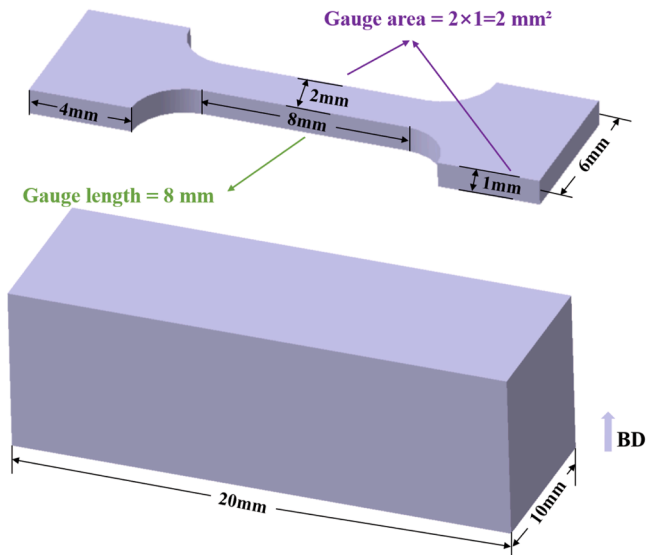


Fig. 10. Schematic diagram of tensile test specimen dimensions.

scanning electron microscopy (SEM, FEI Quanta FEG 450) and electron backscatter diffraction (EBSD, NordlysMax2, UK). EBSD measurements were conducted at an accelerated voltage of 20 kV. The scanning step size values of 0.1 μm and 1 μm were chosen for high and low-magnification EBSD observations, to ensure adequate resolution. EBSD mappings were analyzed using the MATLAB toolbox MTEX [47]. The

samples before SEM observation were first polished with a 90 vol% colloidal silica suspension and a 10 vol% H_2O_2 solution, then samples were etched with Kroll's reagent (85 ML H_2O +5 ML HNO_3 +10 ML HF) for 15 s. Before EBSD characterization, the samples were mechanically polished for one hour to remove the stress.

4.2. Mechanical properties

Fig. 11 (a-c) displays the engineering strain-stress curves of S1, S2, and S3 samples based on the optimized processing parameters. The mechanical properties of all three alloys are remarkably outstanding, and a comparison with their predicted properties obtained by the CIRM-NSGA2 model is shown in Fig. 11 (d). Furthermore, the actual properties of the three optimal alloys are highly consistent with the prediction properties of the CIRM-NSGA2 model (see Fig. 11 (d)). Fig. 11 (e) provides a comparison of the tensile performance of the L-PBF Ti6Al4V alloy among the CIRM-NSGA2 model results and various additive manufacturing and wrought ones. It exhibits a more balanced strength and elongation to fracture compared to most as-built Ti6Al4V in previous works [48–54]. It strongly suggests that the CIRM-NSGA2 model exhibits an exceptional ability to predict the mechanical properties and optimize the process parameters of L-PBF Ti6Al4V alloys.

4.3. Microstructure characterization

Due to its balanced mechanical properties, including yield strength ($950 \text{ MPa} \pm 23 \text{ MPa}$) and elongation to fracture ($11.5 \% \pm 2.0 \%$), the S1 sample is chosen for further in-depth analysis. Fig. 12 shows the microstructural observation of the S1 sample analyzed using OM, SEM,

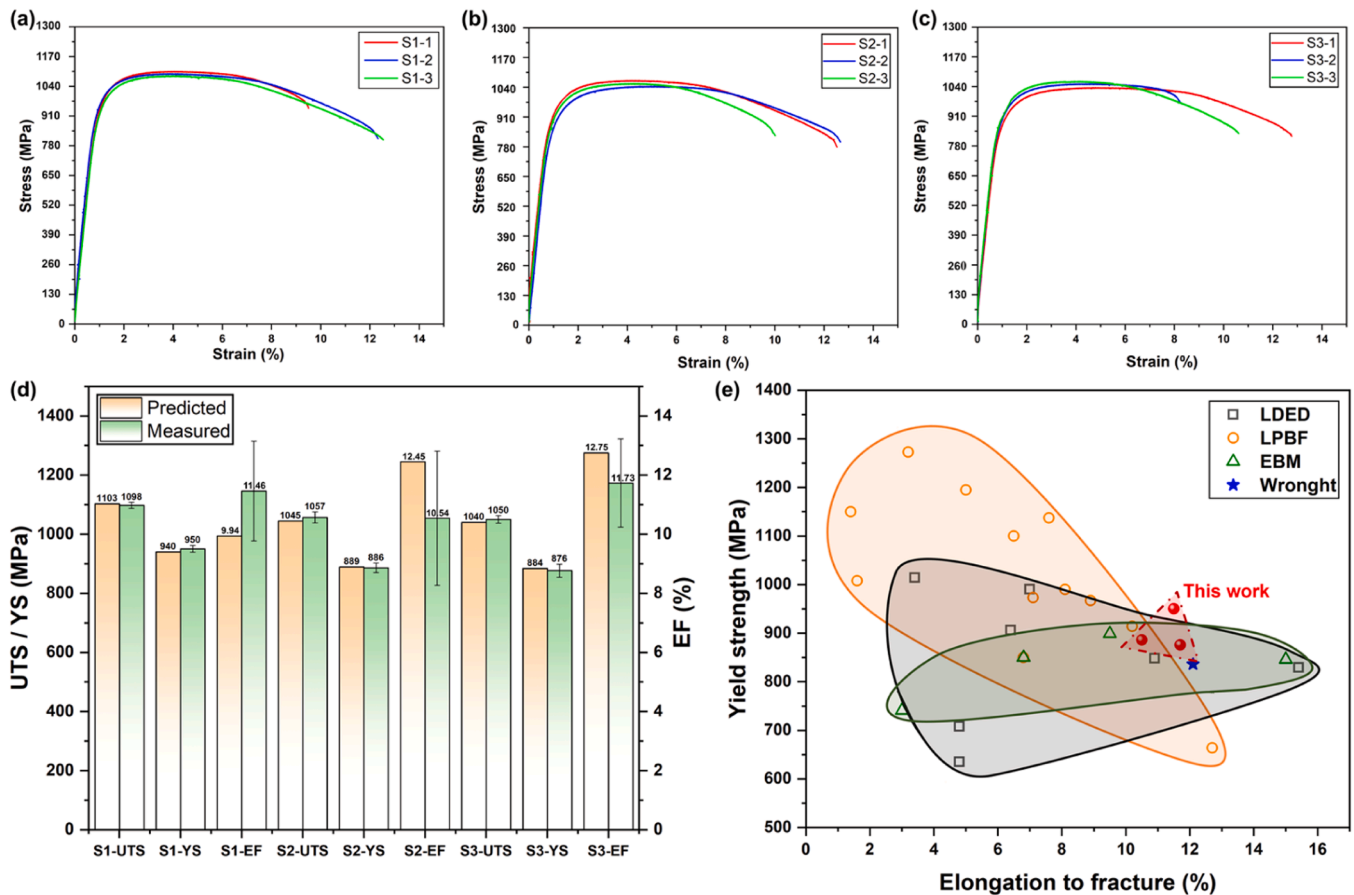


Fig. 11. Stress-strain curves of (a) S1, (b) S2, (c) S3 samples for three experiments; (d) comparison of predicted and measured values of the mechanical properties of S1, S2, and S3 sample; (e) comparison between the yield strength and elongation to fracture of Ti6Al4V alloys obtained by different additive manufacturing and wrought processes [48–60].

and EBSD. From the cross-sectional view (Fig. 12 (a-c)), the large and vertical columnar prior β grains are present in the L-PBF Ti64 alloy, which are growing along the building direction (BD). It is a result of the re-melting of the prior solidified layers during the L-PBF process, which leads to epitaxial and directional solidification[61]. From the SEM image (in Fig. 12 (b)), α' martensite was predominant in the S1 sample, and a few nano-scale dots can also be detected. According to related publications [62,63], such white dots may be considered as the nano-scaled β phase. EBSD analyses were performed on the S1 sample to investigate the microstructural evolution as given in Fig. 12 (c-d). The fine secondary martensite phase between primary ones was predominant [2]. To further reveal the change in size, Table 6 presents the

quantitative data for the thickness of α' martensite. MTEX is employed to perform an ellipsoidal approximation of the martensite and obtain data for its major and minor axes, where the minor axis length is regarded as the thickness of the martensite [2]. According to the thickness variation,

Table 6

Proportion and average thickness of martensite of the sample S1.

Tertiary martensite 0.02 – 0.1 μm (%)	Secondary martensite 0.1 – 1 μm (%)	Primary martensite $\geq$ 1 μm (%)	Average thickness (μm)
1.21	55.22	43.57	1.05 $\pm$ 0.66

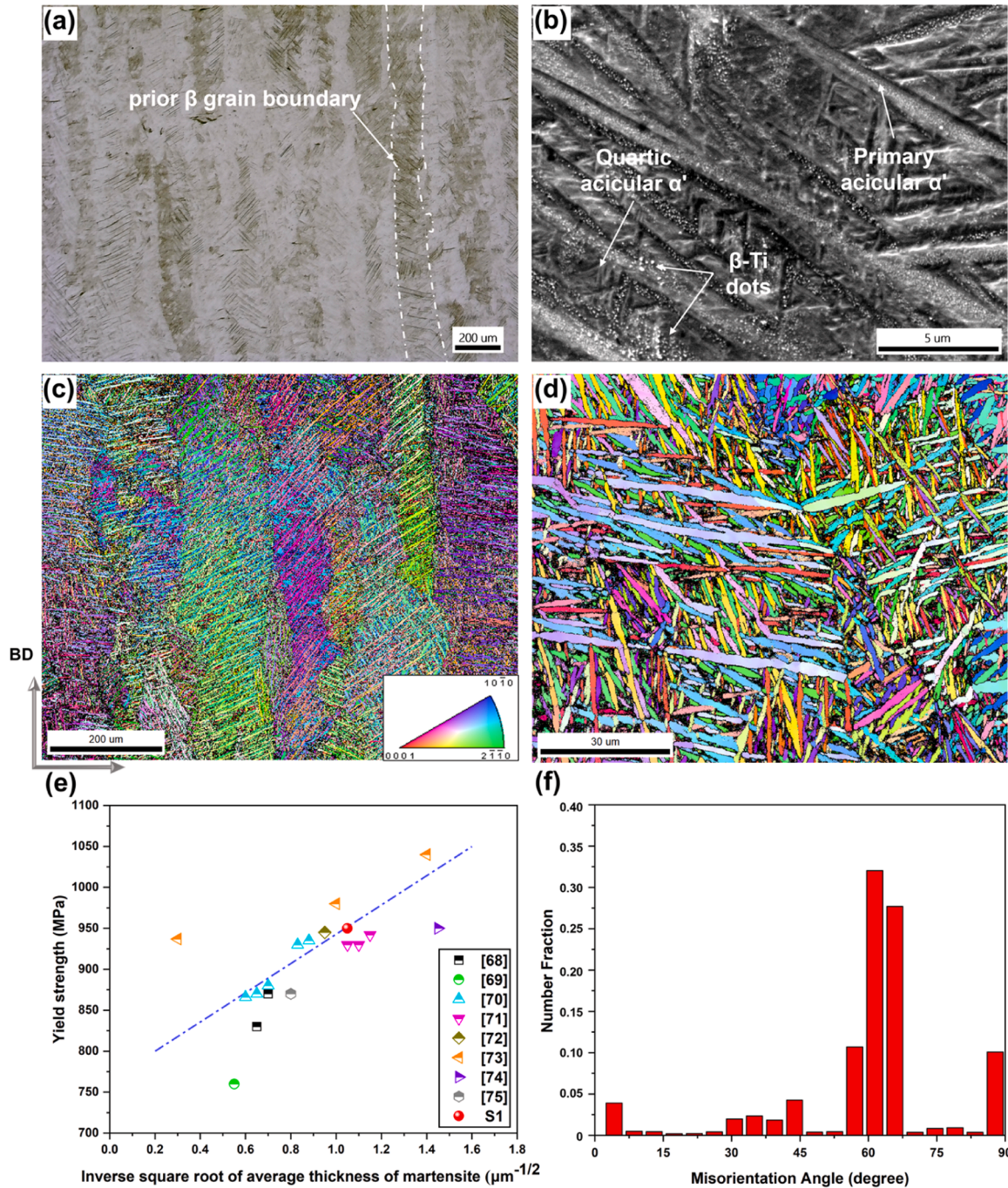


Fig. 12. Microstructural observation of the L-PBF Ti6Al4V alloy: (a) OM (b) SEM image of S1 sample; Inverse pole figures (IPF) at (c) 300 $\times$ magnification and (d) 2000 $\times$ magnification of S1 sample; (e) Comparison of the dependence of yield strength on inverse square root of the average thickness of martensite in different literatures; (f) Misorientations angle distribution histograms of S1 sample.

the observed martensites were classified as primary, secondary, and tertiary ones [64]. Most of the martensite phases possessed thickness values between 0.1 and 1 μm . For Ti6Al4V alloys, the mechanical properties can be assessed using the average thickness of martensite as a characteristic length [65]. Fig. 12 (e) presents a summary of literature data showing the relationship between yield strength and the inverse square root of the average thickness of martensite for Ti6Al4V. The observed relationship is similar to the Hall-Petch relationship [65] found in most of the reported data [66–73]. The yield strength obtained in this study aligns with this relationship.

By observing the grain boundary distribution charts in the low magnification EBSD images (shown in Fig. 12 (f)), it can be found that the proportion of low-angle grain boundaries ($2^\circ \sim 15^\circ$) is about 5 %. Previous studies have shown that a higher proportion of low-angle grain boundaries is beneficial for improving the strength of Ti6Al4V alloy. To improve the elongation to fracture while maintaining the strength, it is appropriate for S1 to have such a proportion of low-angle grain boundaries.

The plastic deformation capability of L-PBF Ti6Al4V alloy is determined by the crystallographic orientation of α' . This study demonstrates the plastic deformation capability of the sample S1 by calculating the Schmid factor (Eq. 8) [74].

$$\text{Schmid factor} = \cos\theta \cdot \cos\lambda \quad (8)$$

θ represents the angle between the loading direction and the slip plane normal, while λ is the angle between the force loading direction and the slip direction.

The basal slip system ($\alpha\{0002\} < 11\bar{2}0 >$) was used with tensile stress acting in the building direction, as the prism and basal slips are the most easily activated modes in titanium alloys [75]. Based on the Elasto-Plastic Self-Consistent (EPSC) model, a critical resolved shear

stress (CRSS) ratio of 1.31 was set for the prism and basal planes [76]. As shown in Fig. 13 (a-b), the proportion of the S1 sample is highest at a Schmid factor of 0.5. Between 0.4 and 0.5, the proportion is 40.25 %, corresponding to the elongation to fracture as compared to previous research [1]. When the Schmid factor accounts for a significant proportion between 0.4 and 0.5, the activation time of the basal system is relatively late, leading to a delayed occurrence of stress concentration, thereby increasing the elongation to fracture of S1. In conjunction with statistical analysis of the width of α' lath in the KAM map (shown in Fig. 13 (c-d)), it was found that higher laser power and higher scanning speed lead to higher KAM values, resulting in a medium proportion of fine α' and a medium dislocation density, which ensures a sufficiently high elongation to fracture for the sample S1.

5. Conclusion

In this work, a combined CIRM-NSGA2 model was developed for predicting the mechanical properties and optimizing process parameters of Ti6Al4V alloy in additive manufacturing. The main conclusions are drawn as follows.

- (1) A clustering integrated regression model (CIRM) has been proposed that combines K-Means clustering algorithm-guided machine learning (ML) regression to establish the relationship between process parameters and mechanical properties. The K-Means algorithm generates three clusters in the model based on the training dataset, and an ML model with the best predictive performance for each cluster is identified. In addition, the SHAP analysis is carried out to interpret the contribution and importance of a feature to the objectives predicted by a specific ML model.

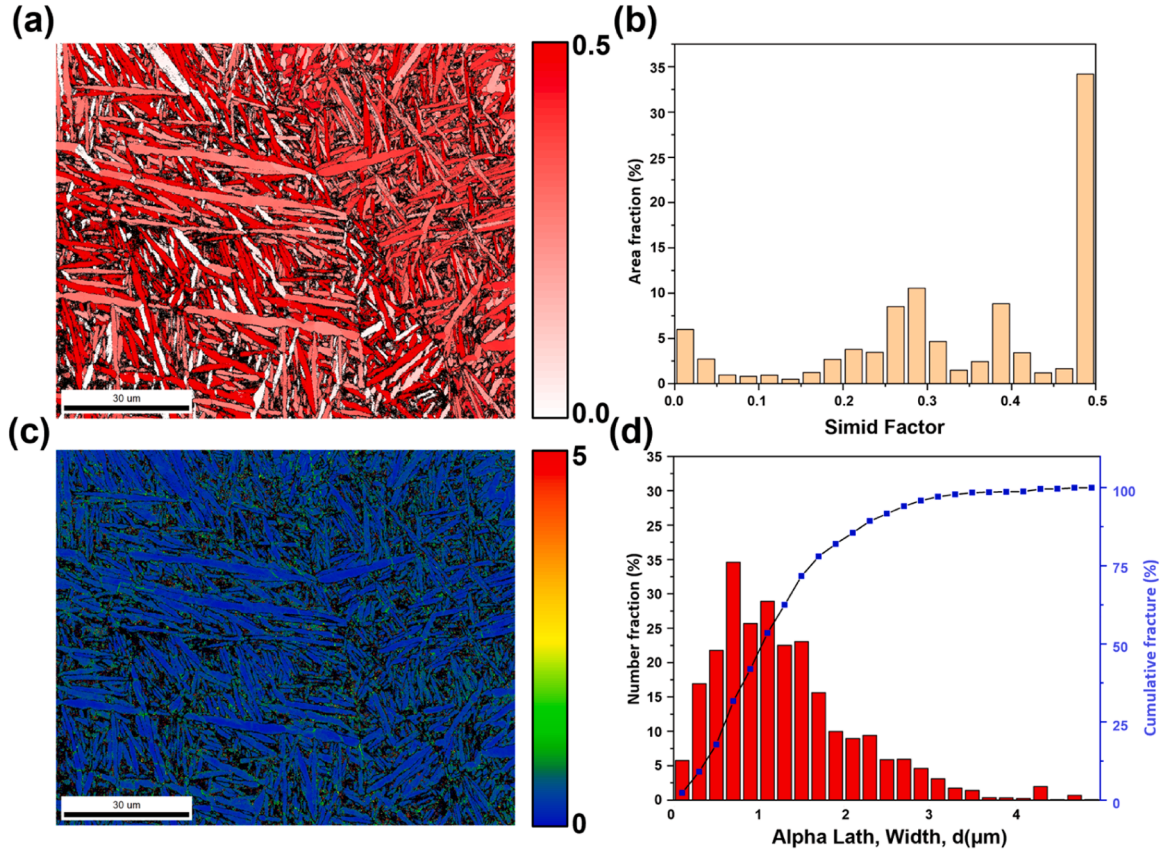


Fig. 13. (a) Schmid Factor map of basal slip system ($\alpha\{0002\} < 11\bar{2}0 >$) corresponding to the S1 sample; (b) Area distribution of Schmid factor on basal slip system; (c) Kernel average misorientation (KAM) maps corresponding to S1 sample; (d) Number Fraction of α lath width corresponding to S1 sample.

- (2) A non-dominated sorting Genetic Algorithm, NSGA2 is employed to determine the optimal solution of process parameters for multi-objective mechanical properties using CIRM as a surrogate model for fast prediction. Two objectives are considered, i.e., YS, and EF, and a Pareto front is produced from the NSGA2. The S1 sample is identified based on the process parameter distributions of data points on the Pareto front.
- (3) Experiments are carried out to validate the selected optimal solution. The results of sample S1 show that the predicted results using CIRM have good agreement with the experimental results indicating the accuracy of the CIRM. In addition, compared to previous work from different literature, the elongation to fracture of sample S1 reaches 11.5 % without sacrificing much strength indicating S1 is more optimal than previous results in obtaining combined better mechanical properties.

As a result, the combined CIRM-NSGA2 optimization model is demonstrated as capable of optimizing the L-PBF process of Ti6Al4V, and it is much more efficient and economical compared to traditional trial-and-error methods. It is promising to be deployed in designing and optimizing different LAM processes for other alloys based on available data.

Code availability

The codes developed for this work are available at <https://github.com/YuHengCAO2000/work-of-master>.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.addma.2024.104341](https://doi.org/10.1016/j.addma.2024.104341).

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